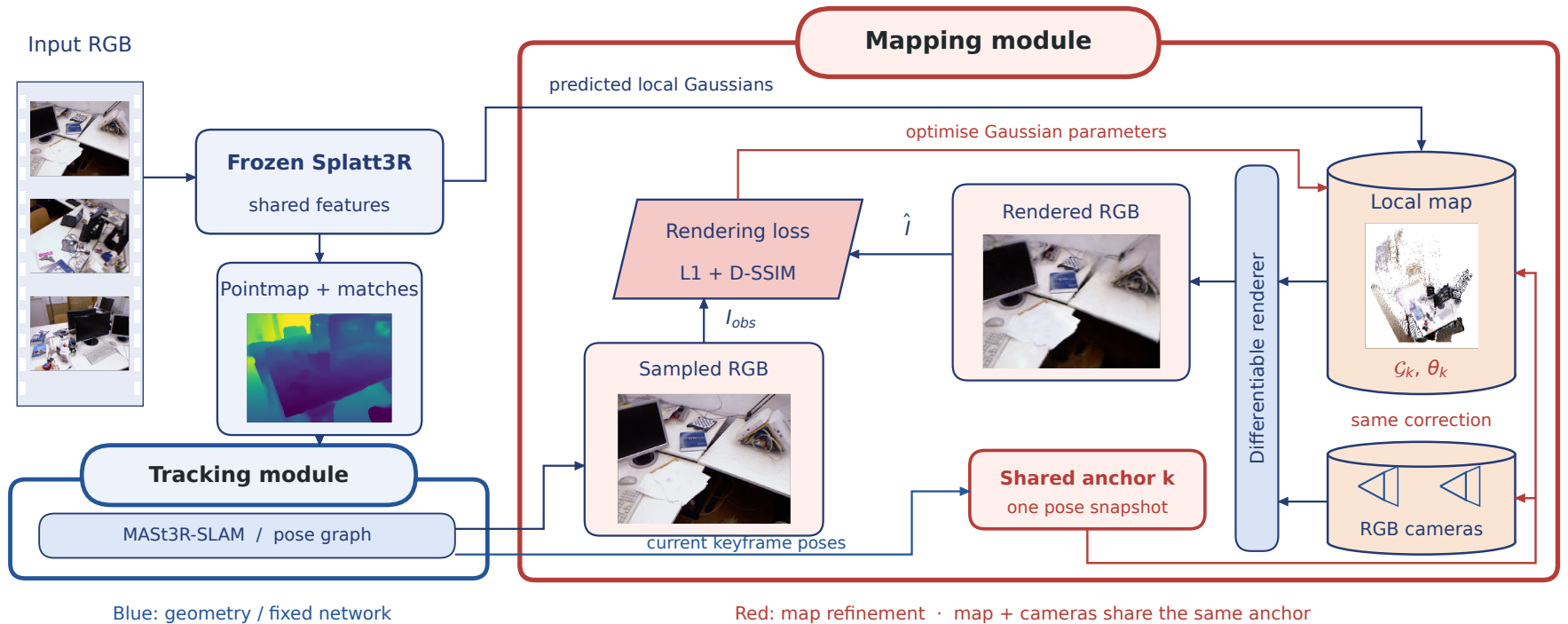

Input RGB
Frozen Splatt3R
shared features
Pointmap + matches
Tracking module
MASt3R-SLAM / pose graph
Mapping module
predicted local Gaussians
optimise Gaussian parameters
Rendering loss
L1 + D-SSIM
$\hat{I}$
Rendered RGB
Local map
$\mathcal{G}_k$, θ_k
Differentiable renderer
I_{obs}
Sampled RGB
current keyframe poses
Shared anchor k
one pose snapshot
same correction
RGB cameras
Blue: geometry / fixed network
Red: map refinement · map + cameras share the same anchor